

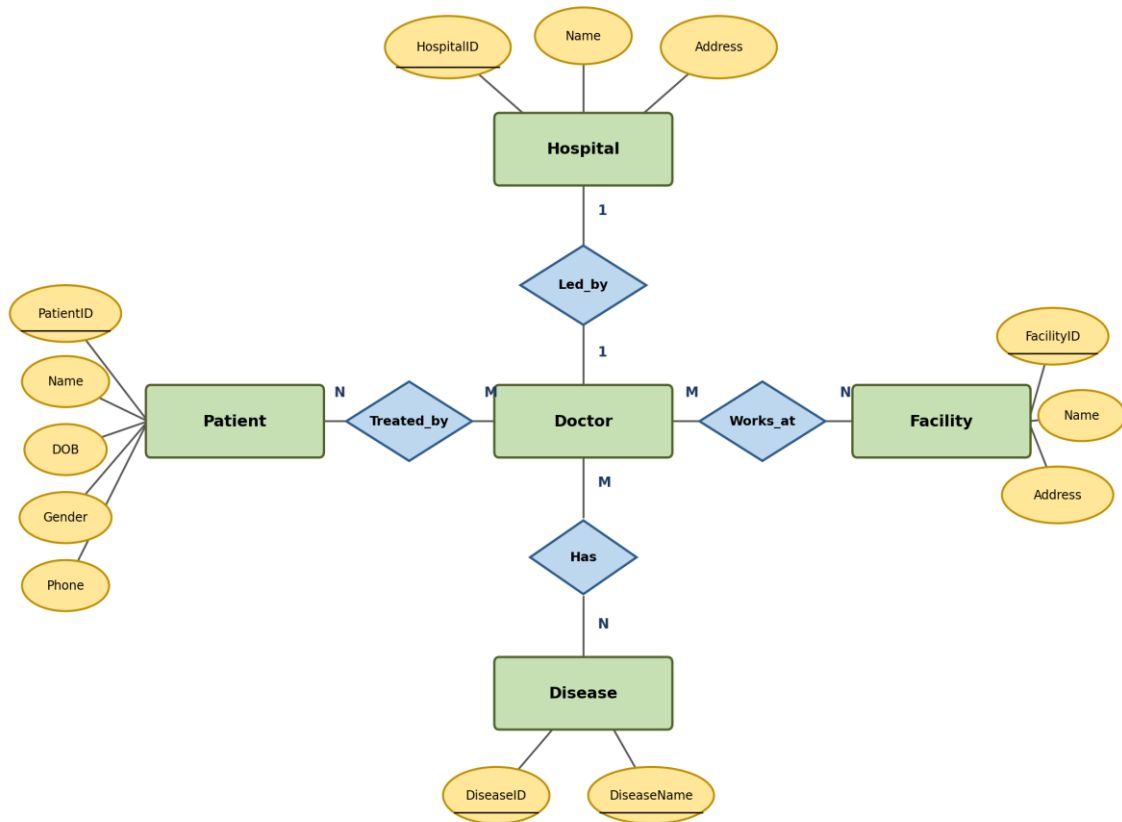
DBMS LAB ASSIGNMENT – 2

Entity–Relationship Modelling & Relational Schema Mapping

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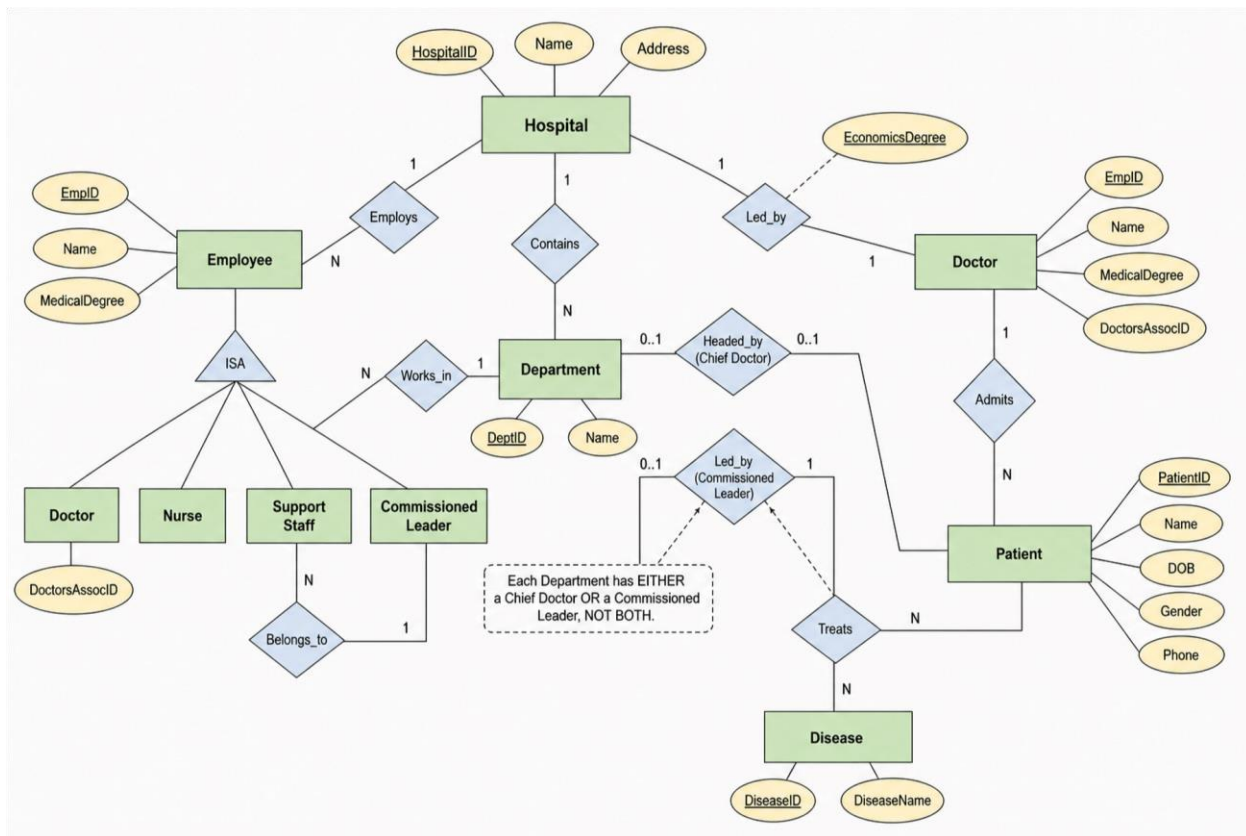
Q1 — Hospital Management ER Diagram

A hospital employs doctors who treat patients for diseases and work at facilities. The entities, attributes, relationships and cardinalities are shown below.



Q2 — Extended Hospital ER Diagram

Employees specialise into Doctors, Nurses, Support Staff and Commissioned Leaders ,Departments are led by either a Chief Doctor or a Commissioned Leader, but not both.



Q3 — Relational Schema Mapping

A (A1, A2, B1)	
Primary Key	A1
Foreign Key	B1 → B(B1)

B (B1, B2, C1)	
Primary Key	B1
Foreign Key	C1 → C(C1)

C (C1, C2, D1)	
Primary Key	C1
Foreign Key	D1 → D(D1)

D (D1, D2, D3, T_A1, S_A1, U_A1)	
Primary Key	D1
Foreign Key	T_A1, S_A1, U_A1 → A(A1)

Q4 — Relational Schema Mapping

1) Basic Mapping

HOSPITAL (HID, HName, DirDegree, DirDoctorID)	
Primary Key	HID
Foreign Key	DirDoctorID → DOCTOR(DoctorID)
Note	<i>DirDoctorID is UNIQUE to ensure one director per hospital.</i>

DOCTOR (DoctorID, DName, Degree, AssocID)	
Primary Key	DoctorID
Foreign Key	— (none)

EMPLOYMENT (DoctorID, HID)	
Primary Key	DoctorID, HID
Foreign Key	DoctorID → DOCTOR(DoctorID) HID → HOSPITAL(HID)

FACILITY (FacilityID, FName, HID)	
Primary Key	FacilityID
Foreign Key	HID → HOSPITAL(HID)

PATIENT (PatientID, PName)	
Primary Key	PatientID
Foreign Key	— (none)

TREATED_AT (PatientID, FacilityID)	
Primary Key	PatientID, FacilityID
Foreign Key	PatientID → PATIENT(PatientID) FacilityID → FACILITY(FacilityID)
Note	<i>Each patient is treated at exactly one facility.</i>

DISEASE (DiseaseID, DName)	
Primary Key	DiseaseID
Foreign Key	— (none)

DOCTOR_PATIENT (DoctorID, PatientID)	
Primary Key	DoctorID, PatientID
Foreign Key	DoctorID → DOCTOR(DoctorID) PatientID → PATIENT(PatientID)
Note	<i>Captures that a patient may be treated by multiple doctors, and a doctor may have multiple patients.</i>

PATIENT_DISEASE (PatientID, DiseaseID)	
Primary Key	PatientID, DiseaseID
Foreign Key	PatientID → PATIENT(PatientID) DiseaseID → DISEASE(DiseaseID)
Note	<i>Captures that a patient may have multiple diseases, and a disease may affect many patients.</i>

2) Extended Mapping (with ISA hierarchy)

HOSPITAL (HID, HName, DirDoctorID)	
Primary Key	HID
Foreign Key	DirDoctorID → DOCTOR(DoctorID)
Note	<i>DirDoctorID is UNIQUE to ensure one director per hospital.</i>

DOCTOR (DoctorID, DName, Degree, AssocID)	
Primary Key	DoctorID
Foreign Key	AssocID → DOCTORS_ASSOCIATION(AssocID)

DOCTORS_ASSOCIATION (AssocID, AssocName)	
Primary Key	AssocID
Foreign Key	— (none)

HOSPITAL_DOCTOR (DoctorID, HID)	
Primary Key	DoctorID, HID
Foreign Key	DoctorID → DOCTOR(DoctorID) HID → HOSPITAL(HID)
Note	<i>A doctor is employed by a maximum of three hospitals — business rule to be enforced.</i>

DEPARTMENT (DeptID, DName, HID, LeaderDoctorID, IsCommissioned)	
Primary Key	DeptID
Foreign Key	HID → HOSPITAL(HID) LeaderDoctorID → DOCTOR(DoctorID)
Note	<i>IsCommissioned: 'Y' if commissioned (not a chief doctor), 'N' otherwise. If IsCommissioned = 'N' then LeaderDoctorID must be a chief doctor.</i>

DEPARTMENT_CHIEF (DeptID, DoctorID)	
Primary Key	DeptID, DoctorID
Foreign Key	DeptID → DEPARTMENT(DeptID) DoctorID → DOCTOR(DoctorID)

Note	<i>Stores all chief doctors of a department. If no record exists for a department, then it has a commissioned leader.</i>
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EMPLOYEE (EmpID, EmpName, Degree)	
Primary Key	EmpID
Foreign Key	— (none)
Note	<i>All employees have a medical degree and are not employed by any other hospital.</i>

DOCTOR_EMPLOYEE (EmpID, AssocID)	
Primary Key	EmpID
Foreign Key	EmpID → EMPLOYEE(EmpID) AssocID → DOCTORS_ASSOCIATION(AssocID)
Note	<i>One-to-one between a doctor (employee) and their doctors' association ID.</i>

NURSE (EmpID, DeptID)	
Primary Key	EmpID
Foreign Key	EmpID → EMPLOYEE(EmpID) DeptID → DEPARTMENT(DeptID)

SUPPORT_STAFF (EmpID, HID)	
Primary Key	EmpID
Foreign Key	EmpID → EMPLOYEE(EmpID) HID → HOSPITAL(HID)
Note	<i>Support staff belongs directly to the hospital.</i>

PATIENT (PatientID, PName, HID)	
Primary Key	PatientID
Foreign Key	HID → HOSPITAL(HID)

PATIENT_DEPARTMENT (PatientID, DeptID)	
Primary Key	PatientID, DeptID

Foreign Key	PatientID → PATIENT(PatientID) DeptID → DEPARTMENT(DeptID)
Note	<i>A patient may be treated in multiple departments until cured.</i>

PATIENT_DISEASE (PatientID, DiseaseID)	
Primary Key	PatientID, DiseaseID
Foreign Key	PatientID → PATIENT(PatientID) DiseaseID → DISEASE(DiseaseID)
Note	<i>During treatment, a patient may be treated for different diseases.</i>

DISEASE (DiseaseID, DName)	
Primary Key	DiseaseID
Foreign Key	— (none)

Q5 — Relational Schema Mapping

E1 (A11, A12, A13)	
Primary Key	(A11, A12)
Foreign Key	— (none)
Constraints	<i>All attributes: NOT NULL</i>

E2 (A21, A22, E1_A11, E1_A12)	
Primary Key	A21
Foreign Key	(E1_A11, E1_A12) → E1(A11, A12)
Constraints	<i>All attributes: NOT NULL</i>

E3 (A31, R4_A11, R4_A12, R5_A11, R5_A12, AR4)	
Primary Key	A31
Foreign Key	(R4_A11, R4_A12) → E1(A11, A12) (R5_A11, R5_A12) → E1(A11, A12)
Constraints	<i>A31, AR4: NOT NULL. R4_A11, R4_A12, R5_A11, R5_A12: NULL allowed.</i>

E4 (A41, E2_A21, E3_A31)	
Primary Key	A41
Foreign Key	E2_A21 → E2(A21) E3_A31 → E3(A31)
Constraints	<i>All attributes: NOT NULL</i>

AR4 (AR4)	
Primary Key	AR4
Foreign Key	— (none)
Constraints	<i>All attributes: NOT NULL</i>